

- KT Add something about appropriate module selection

Module 1 Lecture Outline: AI in Your World – Seeing the Invisible

Hello and welcome to TEK 100! I'm Tama Thé

And I'm Katie Thompson. Welcome to Module 1! AI in your World - Seeing the Invisible

Introduce core theme of the module:

AI is the buzziest buzzword in the news, in society, in business, in medicine. It's pretty much everywhere. The core theme of this module is to understand what AI actually means, beyond the hype and the headlines. We're going to learn to see the invisible systems that are already shaping your world, from your social media feeds to the way you study for an exam. By the end of this module, you won't just be a passive user of AI; you'll be a critical observer, equipped to ask the important questions about how these tools work, where they're being used, and whether they're fair.

Tama, you built this module for us, tell us about it.

In this first module, we're going to build a foundation by understanding the core principles of AI. This is a completely no-code course. While we will touch on some technical concepts it's only to provide the context you need to understand how these systems work and why they produce the results they do. Our entire focus is on equipping you with the practical skills to use artificial intelligence like a pro.

In this module, we'll begin with videos from our experts to explore what AI is and why learning about it now is so critical.

Then, we'll get hands-on by setting up free accounts on the major AI platforms—Gemini, ChatGPT, CoPilot, and Claude. We will use these tools throughout the course period. Some of them have free trials that are open to students.

Next we'll get into some basic AI vocabulary.

And for that, I've designed a custom GPT that you'll use to explain the more complicated terms, like neural networks, in a context that you would prefer. Like how Taylor Swift building a concert setlist will use specifically chosen songs to create a specific audience reaction is a lot like how AI researchers train models using specific weights to create a specific outcome.

Once you're done with that, join back up with us in the discussion board and we can talk about how AI pattern recognition is like a Pokédex for Pokémon.

Finally, we'll close the module by looking ahead to your final project, starting to brainstorm how you can use AI to solve a real problem in a community you care about.

Begin opening lecture

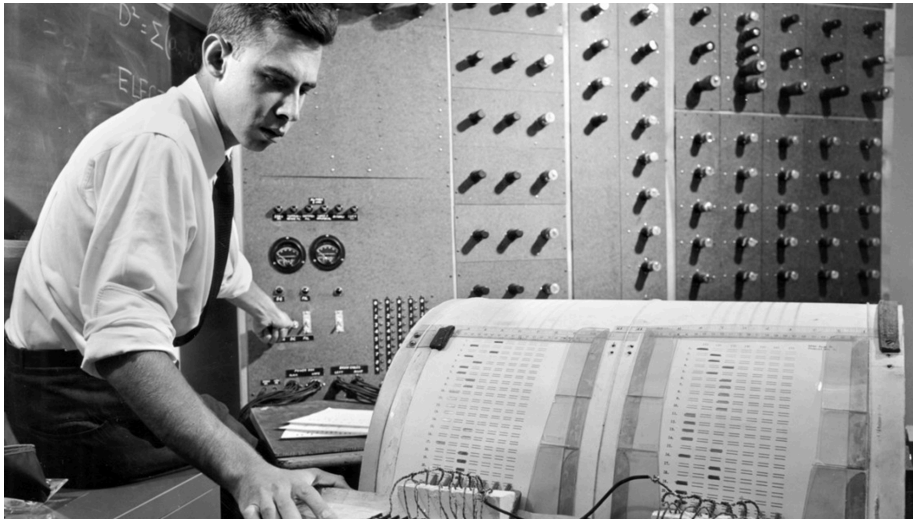
Without any further ado, What is AI?

AI for specific tasks (pattern recognition, prediction). It learns from examples, not explicit instructions or code.

The "Why": AI manages the world's data explosion at a speed and scale humans cannot.

Old vs new AI

AI has been around for a long, long time. This is a picture of Frank Rosenblatt and his perceptron. This was taken in 1954, and the term AI was invented around 1956. This is most likely the first actual artificial intelligence machine that was designed to recognize patterns.

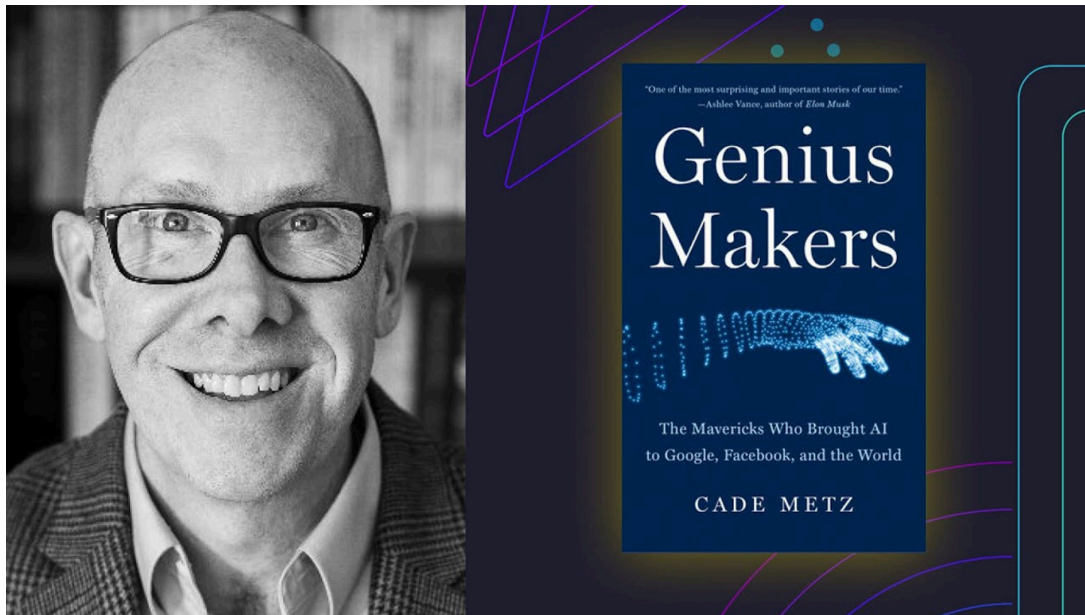


This is a picture of Jeffrey Hinton, who won the Nobel Prize last year in 2024 for his discovery of backpropagation, which significantly advanced our ability to do sophisticated machine learning.



And that's it! That was 75 years of the history of Artificial Intelligence!!

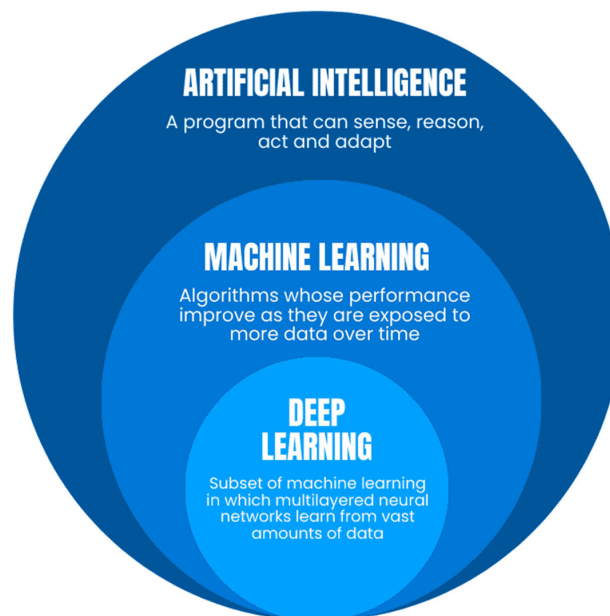
The rest of it is beyond the scope of this course, but it really is a fascinating history. If you want more, you should take Dr. Brent Harrison's introductory course on AI, or please read this book, "Genius Makers" by Cade Metz. It was written and published before ChatGPT and large language models came out, but it is an excellent recounting of the history of AI up until this current explosion.



How is AI used?

Predictive modeling, GenAI tools, AI agents

Pie chart of ChatGPT proportion vs. everything else



The reason artificial intelligence has become such a talked about concept is because of its use in generative AI. Generative AI Can track its origin back to this paper published in 2017 by the team at Google Brain. "Attention is all you need." This is what created the transformer technology. Essentially,

older technology was only really able to make predictions based on the previous few inputs. The transformer technology was able to use a large quantity of language in context. So now, instead of just providing you with basic predictions, these AI systems can use language in ways that mimic the creativity in the way we talk and write.

We're going to talk more about the science behind generative AI later in the module, because understanding how they work will help us use these tools appropriately and predict when these models break down.

AI in Your World:

In Your Pocket: Recommendation algorithms (Netflix, TikTok), writing assistants (Grammarly), and brainstorming tools (ChatGPT).

These are all forms of Artificial Narrow Intelligence. These are AI systems that are designed to do one single job but do them exceedingly well.

This contrasts with Artificial General Intelligence. This is the speculative future where AI is so smart that it's able to design systems that makes itself smarter. To the point that it can eventually do anything. Because even the experts designing AI systems can't agree on when or what this will look like, This is also outside of the scope of this course.

On Campus: For the past 2 months, we have been tracking down people all across campus who are using AI in different ways. From people at the Center for Applied AI who are building sophisticated fine-tuned language models for the people of Kentucky to the leaders of the college of social work who are Using artificial intelligence to make social workers in Kentucky more effective and provide better service to people and children.

Some of the other major ways AI is being woven into the fabric of our lives are in

logistics and the supply chain: UPS uses AI to plot routes, saving time, gasoline, and improving delivery times, while Amazon has a predictive model to forecast demand.

In Healthcare AI is interpreting Xrays, monitoring patients, and recognizing

patterns of disease years, even decades before physicians would be able to pick them up.

The financial sector uses AI to detect fraud

My cellphone carrier is pretty good at guessing when I'm getting a spam call and screens those out

Self driving cars, weather mapping, apparently there's even AI in my Axe Body spray



As we designed this course, we also wanted to make sure that we were addressing some of the concerns and hesitations that people have about AI. These systems are incredibly powerful, and to be responsible with them, we need to keep a critical eye on AI. To that end, here are a couple of recurring threads about some of the ways we need to remain cautious when using AI:

Myths

Myth: AI will replace jobs. **Reality:** It's a tool for *augmentation*, freeing humans for tasks requiring creativity, empathy, and critical thinking.

Myth: AI is objective and always accurate. **Reality:** AI reflects the biases in its training data (the "cookbook") and is prone to making factual errors ("hallucinations").

Myth: Using AI is always cheating. **Reality:** The line is defined by transparency and intent. It can be a legitimate tool for brainstorming, outlining, and editing when used responsibly. Afterall, spell check is an AI tool...

Myth: AI is only for tech-savvy people and data scientists. Many modern AI tools are designed with user-friendly interfaces. From search engines to email assistants, most people already use AI daily without realizing it. New platforms are increasingly designed for non-technical users.

Katie, how do you use AI?

Formatting paperwork to let me spend more time interaction with students, collaborators, faculty, colleagues, staff

Preparing for meetings with new or different perspectives

Module Takeaway

- Build foundational AI awareness.
- Equip students to recognize, understand, and question AI's impact on their world.

That's a lot, let's get in to our first videos

Lecture 2

This course is about ideas. It's about understanding a technology so that you can be an informed, critical, and confident citizen in that world.

To do that, we first need to be able to talk to each other. Before we can have a meaningful conversation about anything—whether it's film, sports, or science—we need a shared language. That's our only goal today: to build a small toolkit of foundational words so we can all participate in the conversation about AI.

I promise that not only will we never test you on the details of Neural networks or the transformer technology that went into ChatGPT, We will probably never even bring it up again. But, I think to really understand how this technology works, we need to have just a little bit of context.

Let's start with what is AI?

Artificial Intelligence (AI)

Definition: Artificial Intelligence, or AI, is a broad field of science focused on creating technologies that allow computers to do tasks that would normally require human intelligence. This includes things like understanding spoken language, recognizing faces in photos, analyzing data, and making recommendations.

When you ask Siri or Alexa for the weather, you're interacting with an AI.

It's like the software that powers your robot vacuum. - Minesweeper story

You can think of AI less like a robot brain and more like a **pattern-finding assistant**.

2. Algorithm

Definition: An algorithm is simply a step-by-step set of instructions for a computer to follow to accomplish a specific task. It's a finite, logical process that, if followed correctly, will produce a consistent result every time.

Example: Think about a GPS app like Google Maps or Waze. When you ask for

directions, the app runs an algorithm. The instructions might look something like this: Step 1: Check for current traffic jams. Step 2: Consider any reported road closures. Step 3: Calculate the distance of all possible routes. Step 4: Present the route with the shortest estimated travel time.

Analogy: The simplest and best analogy for an algorithm is a **recipe for baking cookies**. The recipe gives you a list of ingredients (that's the data) and a clear, step-by-step set of instructions to follow. If you follow the recipe, you'll get cookies. The recipe itself isn't the cookies, it's just the instructions for *how* to make them.

Not all algorithms are AI. For instance, in algebra, we learned PEMDAS. This is an algorithm that we all use to make sure that we all get the same answer to the following equation: $(3 * 2 + 1)^2$

3. Model

Definition: An AI model is the final product that's created after an algorithm has processed, or "learned from," a set of data. The model is the specific program that can then make predictions, classifications, or decisions on new, unseen data. GPT 5 is an example of a model, while ChatGPT is an example of a chat program that allows us to access the model easily.

Example: Your email's spam filter is a perfect example of a model. Engineers took an algorithm and "fed" it millions of emails, with each one labeled as either "spam" or "not spam." The result of that process is a trained "spam detection model." Now, that model can look at a brand-new email it has never seen before and make a very accurate prediction about whether it's junk.

4. Machine Learning

Definition: Machine learning is the process that allows a computer to learn from data and find patterns on its own, so it can improve its performance over time without humans having to write explicit, step-by-step rules for every single

possibility. It's how we get from the algorithm to the model.

Example: Think about the photos app on your phone that automatically recognizes and groups pictures of the same person. You never told your phone, "Look for eyes that are this shape and a nose that is this size." Instead, through machine learning, the app analyzed thousands of your photos and learned on its own what makes your friend's face unique.

Analogy: Machine learning works by "learning from example," just like we do as kids. No one gives a toddler a complex, technical algorithm for what a "dog" is. We just show them lots of examples: "Look, that's a dog. That poodle is also a dog. Oh, but that's a cat." Over time, the child's brain builds its own internal "model" of what constitutes a dog. Machine learning is that same process, but for computers.

5. Training Data - Needs updating - Reddit/4chan/legal concerns?

Definition: Training data is the information—whether it's text, images, sounds, or numbers—that is used to teach, or "train," an AI model during the machine learning process.

Example: To create a powerful language model like ChatGPT, it was trained on an enormous dataset consisting of a huge portion of the public internet. This included countless websites, digital books, articles, and other text data. That massive collection of information is the training data that taught it grammar, facts, different writing styles, and how to have a conversation.

Analogy: Training data is like all the **textbooks, lecture notes, and practice problems you give to a student** who is preparing for a huge final exam. The more high-quality and diverse the study material (the training data), the better the student (the AI model) will be able to understand the subject and answer questions on the test.

6. Garbage In, Garbage Out - Update

Definition: "Garbage In, Garbage Out," is one of the most fundamental concepts in all of computer science. It means that if you put flawed, biased, or poor-

quality ("garbage") data into a system, you will get flawed, poor-quality, or "garbage" results out of it.

Example: Imagine you want to build an AI model to predict house prices. If you only train it on data from multi-million dollar mansions in Beverly Hills, and then ask it to predict the price of a small apartment in Lexington, its prediction will be completely wrong—it will be garbage. The input data didn't represent the real world, so the output can't be trusted.

Analogy: An AI model is like a **blender**. The blender is a powerful tool, but it can't fix bad ingredients. If you put in fresh, delicious fruit, you get a wonderful smoothie. But if you put in rotten, moldy fruit, you're going to get something you definitely don't want to drink. Garbage in, garbage out.

7. Neural Network -

Definition: A Neural Network is the underlying architecture of most modern AI, loosely inspired by the structure of the human brain. It consists of interconnected layers of nodes, or "neurons," that process information, recognize patterns, and pass signals to one another.

When neural networks are combined into layers, it allows these models to do more Deep learning. This is a type of machine learning with many layers. When people talk about AI being a black box, this is why. While we can design these neural networks and control the training data, we don't actually understand how the model is creating the output it is.

8. Generative AI (GenAI)

Definition: Generative AI is a class of artificial intelligence systems capable of creating new, original content—such as text, images, music, or code—rather than just analyzing or classifying existing data.

Example: When you use a tool like Midjourney to create an image from a text description, or use ChatGPT to write a poem, you are using Generative AI. These tools aren't just finding existing images or poems; they are generating entirely new ones based on the patterns they learned from their training.

Analogy: Generative AI is like a **creative chef who has studied thousands of recipes**. You can give the chef a few ingredients (your prompt), and instead of just finding a recipe that uses them, the chef creates an entirely new and unique dish just for you, based on everything they know about how flavors and techniques work together.

- Top Chef example
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6. Large Language Model (LLM)

Definition: A Large Language Model is the specific technology that powers generative text AI like ChatGPT. It is an AI model that has been trained on an immense corpus of text data, enabling it to understand, summarize, predict, and generate human-like language.

Example: The technology behind ChatGPT is a prime example of an LLM. It was trained on a vast amount of text from the internet, allowing it to have conversations, answer questions, and write different kinds of text by predicting the next most likely word in a sequence.

Analogy: An LLM is like a **librarian who has read every book in the world**. They haven't just memorized the books, they've learned the connections between words, ideas, and writing styles. Because of this, they can answer almost any question, write an essay on any topic, or even mimic the writing style of a famous author.

10. Bias

Definition: AI bias occurs when an AI system produces results that are systematically prejudiced. This happens when the training data reflects existing human biases and stereotypes, which the AI then learns and often amplifies.

Example: A few years ago, Amazon had to scrap an AI recruiting tool they were building. They discovered it was biased against female candidates. Why? Because the model was trained on the company's hiring data from the previous 10 years, and most of the applicants during that time had been men. The AI "learned" that male candidates were preferable and started penalizing resumes that contained words like "women's," as in "captain of the women's chess club".

Analogy: A biased AI is like a **curved mirror at a carnival funhouse**. It reflects the world it was shown in its training data, but it distorts that reflection. It might exaggerate some features, shrink others, and present a skewed version of reality. It's not reflecting the world as it *should be*, but rather reflecting the biases that already exist in our society, and sometimes making them even more pronounced.

- Example: 2D vs. 3D map

12. Prompt

Definition: A prompt is the specific instruction, question, or input given by a human user to an AI model to elicit a response. Crafting effective prompts is a core skill for a co-intelligence partnership.

Example: Instead of just typing "write about dogs," a better prompt would be: "Write a 300-word blog post for a new dog owner about the importance of positive reinforcement training for puppies. Use a friendly and encouraging tone".

Analogy: A prompt is like **ordering a drink from a barista**. If you just say "coffee," you'll get something generic. But if you give a clear, detailed order specifying the size, type, milk, and flavor, you'll get exactly what you want. The

quality of your instructions directly determines the quality of the result.

11. Noise

Definition: In AI, noise refers to any irrelevant, random, or erroneous data within a dataset that can confuse a model and make it harder for it to detect the true, meaningful patterns—what we call the "signal".

Example: Imagine you're training an AI to understand customer sentiment by analyzing thousands of product reviews. Some of those reviews might contain typos, some might have random keyboard mashes like "asdfghjkl," and some might even be for the wrong product entirely. All of this incorrect and irrelevant information is noise, and it can make it much harder for the model to learn the true signal of what customers actually think.

Analogy: Trying to find the signal in noisy data is like trying to listen to your favorite song on an old AM radio station while driving through a tunnel. The song itself is the **signal**—the meaningful information you want. The crackling and static is the **noise**—the random interference that gets in the way and makes it hard to hear the signal clearly.

Goal in data science is not to remove the noise from the data, but to understand how to work with noisy data.

9. Hallucination

Definition: An AI hallucination is when a generative AI model produces an output that is confident and plausible-sounding, but is factually incorrect, nonsensical, or completely fabricated and not based on its training data.

Example: This is a famous real-world example. A lawyer used ChatGPT to help write a legal brief for a court case. The AI invented several completely fake court cases to use as precedents, complete with very official-looking but non-existent citations. The lawyer submitted the brief without checking, and the judge, unable to find any of the cited cases, fined the lawyer and his firm for

submitting fabricated information.

Analogy: An AI hallucination is like an **overconfident student taking an exam**.

The student gets to a question they don't know the answer to, but they don't want to leave it blank. So instead of admitting uncertainty, they make up a very detailed, elaborate, and confident-sounding answer, hoping it's close enough to be right. It might look good at first glance, but it's completely fabricated.

13. Human in the Loop (HITL)

Definition: Human in the Loop, or HITL, is a system where a human being is intentionally placed into the AI's process to **supervise, correct, or validate** its decisions. It's especially important in high-stakes situations where errors could have serious consequences. This is our first line of defense for ensuring AI is used responsibly.

Example: On social media platforms, an AI might automatically flag a post for potentially violating community guidelines. But before the post is taken down or a user is banned, a human content moderator—the human in the loop—reviews the AI's decision to apply real-world context and nuance, preventing errors and unfair censorship. Similarly, doctors use AI to help identify potential tumors in medical scans, but it is always the human doctor who makes the final diagnosis.

Analogy: A Human-in-the-Loop system is like a **student driver's car that also has a brake pedal on the instructor's side**. The student (the AI) is doing most of the driving. But the instructor (the human) is in the loop, constantly monitoring and ready to hit the brake or grab the steering wheel if the student makes a mistake or encounters a complex situation they can't handle. It's all about safety, oversight, and ultimate human responsibility.

- **Example: Vacuum Robot (human in the loop, but human not ever doing the vacuuming)**
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14. Collaborative Intelligence

Definition: Collaborative Intelligence is the idea of a **true partnership between humans and AI**. It's a collaborative synergy where the unique strengths of both human and machine intelligence are combined to achieve results that neither could accomplish alone. It's the next step beyond just supervision; it's about creation.

Example: A marketing team needs to come up with a new ad campaign. They could spend days brainstorming a few ideas. Or, they **could use an AI to generate a hundred different concepts, slogans, and images** in just a few minutes. The human experts then use their creativity, experience, and knowledge of their audience to select the top three ideas and refine them into a brilliant final campaign. The AI provides the scale and speed; the humans provide the strategic wisdom and taste.

Analogy: The best analogy for collaborative intelligence is a **pilot and a co-pilot flying a modern airliner**. The AI co-pilot can process massive amounts of flight data, manage incredibly complex systems, and fly the plane with superhuman precision. The human pilot provides the strategic direction, handles unexpected turbulence, communicates with air traffic control, and makes the final, critical